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CMi EPFL Center of MicroNanoTechnology

☑ Semestral Project ☐ Master Project ☐ Thesis ☐ Other

Si-Nano Wire Reconfigurable FET (Si-NW RFET) Microfluidics

Description of the fabrication project

The Si-NWs are manufactured in a top-down approach. Two Al reservoirs will be deposited at the two extremities of the Si-NWs, to form Al/Si Schottky barrier. Then they are annealed in forming gas in a furnace to allow the Al/Si diffuse into the Si-NWs[2]. The resulting Si-NWs will be oxidated to form an Sio2 layer using dry oxidation to induce a radial compressive strain into the silicon region[3]. Three electrodes are then deposited on the Si-NWs at the two Schottky barrier and the Si channel as program gates and control gate. The resulting device can be switched between p-type FET and n-type FET depending on the biased voltage on the program and control gate. This allows the FET to probe both negative and positive biomarkers depending on the requirements.

Here we write the steps needed to be done on CMi for the creation of a mask for the pattern of the microfluidic design.

Technologies used *!! remove non-used !!*			
E-beam lithography, PDMS line			
Ebeam litho data - Photolitho masks - Laser direct write data			
Mask #	**Critical Dimension**	**Critical Alignment**	**Remarks**
M1	**300um**	>1um	Photolithography
Substrate Type			
Chrome Mask 5’’			

Interconnections and packaging of final device

Thinning/grinding/polishing of the samples is required at some stage of the process.

☑ No ☐ Yes => confirm involved materials with CMi staff

Dicing of the samples is required at some stage of the process.

☐ No ☑ Yes => confirm dicing layout with CMi staff

Wire-bonding of dies, with glob-top protection, is required at the end of the process.

☑ No ☐ Yes => confirm pads design (size, pitch) and involved materials with CMi staff

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Step-by-step process outline

Step	Process description	Cross-section after process
1	**Description:** Photolithography **Material:** Cr Mask AZ1512 (530nm) **Machine:** VPG200 (Z5) **Remark:**	
2	**Description:** Mask development and etching **Machine:** Hamatech (Z6) **Remark:**	